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Cluster assignment indicates four distinct homegarden types in southwest Uganda.

The Greater Bushenyi region in southwest Uganda is characterized by high ecological and cultural diversity, but also by high levels of poverty and malnutrition. Much of the natural vegetation (Guineo-Congolian rainforest) has been transformed into banana-based homegardens (HG), the primary source of food for local people. Inventories of 102 HGs aimed to define HG types in relation to their level of agrobiodiversity as a first step in conceptualizing interventions to help address regional food insecurity. Ward's minimum variance hierarchical cluster analysis, (with Mahalanobis distance measure and squared Euclidean distances of $\log(e)$ transformed individual densities of useful plant species per 1000 m² HG area) divided HGs into four distinct HG floristic diversity types. Clusters 1 (n=39 HGs) and 2 (n=10) had the greatest Shannon H' and greatest Pielou's J'. Cluster-3 (n=10) had the highest species richness, but only medium H' and J'. Cluster-4 (n=43) had the greatest abundance and richness of annual and perennial herbs. Clusters 3 and 4 both had the greatest abundance and density of plants per 1,000 m². Linear discriminant analysis (LDA) revealed 98% correct classification and a Breiman's random forest model explained 51% of variance. Models of the cluster assignment and plant species densities indicated 13 plant species important to the clustering model. Mean squared error (MSE) values in the random forests model were strongest for eight species *Phaseolus vulgaris*, *Coffea arabica*, *Amaranthus dubius*, *Amaranthus hybridus*, *Saccharum officinarum*, *Amaranthus spinosus*, *Solanum torvum*, and *Artocarpus heterophyllus* (MSE<1.3%). Stepwise variable selection indicated 11 species significantly minimizing the Wilk's λ ; *A. spinosus*, *P. vulgaris*, *C.*

arabica, *A. hybridus*, *S. torvum*, *Gynandropsis gynandra*, *Hibiscus acetosella*, *Gynura scandens*, *Grevillea robusta*, *A. dubius*, and *Antiaris toxicaria* ($p < 0.001$). Increasing the nutritional value of these food systems may begin with a diverse approach based on the identified HG types and important plants.

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Keywords:

food security

agrobiodiversity

homegarden

cluster analysis

Mahalanobis distance

Euclidean distance

richness

abundance

indigenous knowledge

Coffea arabica

Amaranthus dubius

Amaranthus hybridus

Saccharum officinarum

Amaranthus spinosus

Solanum torvum

Artocarpus heterophyllus

Gynandropsis gynandra

Hibiscus acetosella

Gynura scandens

Grevillea robusta

Antiaris toxicaria

Shannon H'

Pielou's J'

Uganda.

Presentation Type: Poster

Number:

Abstract ID:50

Candidate for Awards:Julia F. Morton Award
